

FinWise AI: Intelligent Loan Eligibility, Credit Analysis & Financial Advisory Platform

Project Description:

FinWise AI is a premium, high-fidelity BFSI (Banking, Financial Services & Insurance) web platform designed to simplify personal financial diagnostics. The platform gives users instant access to four integrated financial tools — a Loan Eligibility Checker, a Credit Score Analyzer, a Quick EMI Calculator, and an AI Financial Advisor — all within a single, seamless glassmorphism-styled interface.

Built entirely with vanilla HTML, CSS, and JavaScript and powered by Anthropic's Claude API (model claude-3-5-sonnet-20241022), FinWise AI evaluates applicant salary, credit score, existing liabilities, and age against predefined BFSI business rules to instantly determine loan eligibility and risk category. The Credit Score Analyzer applies a FICO-style weighted scoring model across five factors, the EMI Calculator uses the standard reducing-balance formula to generate a full month-by-month amortization schedule, and the AI Financial Advisor uses Claude to deliver contextual, chat-based financial coaching based on the user's live diagnostic profile.

The platform operates in two modes — a Demo Mode that runs entirely offline using simulated logic, and a Live Mode that calls the Claude API directly from the browser. All diagnostic data is persisted client-side using localStorage, with an optional Google Sheets webhook (via Google Apps Script) for centralized data logging and backup.

Scenarios

Scenario 1: Salaried Applicant Approval. An applicant named “chsa” with a monthly salary of ₹2,00,000, a CIBIL score of 750, an existing EMI of ₹2,000, and age 22 submits the Eligibility Criteria Form. All four business rules pass (Salary $\geq$ ₹20,000, Score $\geq$ 600, Existing EMI $<$ ₹20,000, Age $\geq$ 21), and the Diagnostics Ledger instantly returns “Loan Approved” with a Low risk level and an eligible loan amount of ₹40,00,000.

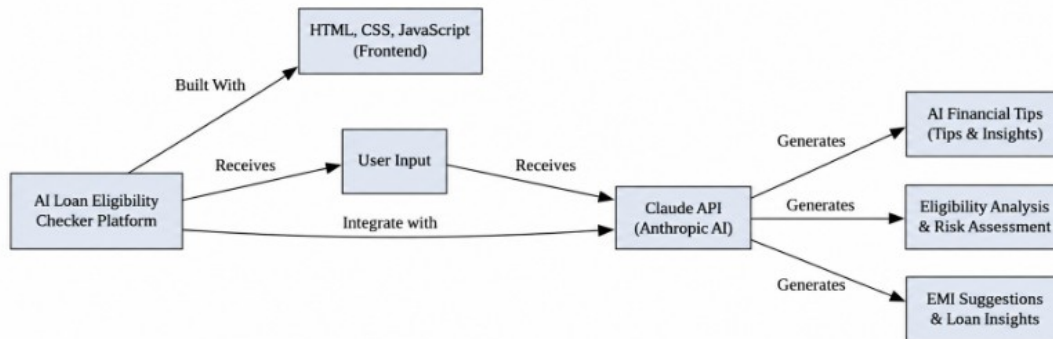
Scenario 2: Credit Score Estimation. A user with no loan defaults in the last two years, a 25% credit utilization ratio, 5 years of credit history, 2 active credit types, and 1 hard inquiry in the last 6 months runs the Credit Risk Calculator. The FICO-style engine computes a weighted score across the five factors (Payment History, Utilization, Credit Age, Inquiries, Diversity) and returns an estimated score, risk tier, and a factor-wise breakdown in the Credit Diagnostics panel.

Scenario 3: EMI Planning. A user planning to take a loan enters the Loan Principal Amount, Annual Interest Rate, and Tenure (months) into the Quick EMI Calculator. The engine applies the reducing-balance EMI formula and instantly renders the monthly installment along with a complete Payment Schedule showing the interest, principal, and balance for every month of the tenure.

Scenario 4: AI-Guided Financial Coaching. After completing an eligibility check (Income ₹2,00,000/mo, Credit Score 750, Status Approved, Requested Loan ₹40,00,000), the user opens the AI Financial Advisor. Claude automatically initializes the diagnostic profile, greets the user by name, and offers proactive budgeting guidance.

The user can then chat freely or click “Generate Audit Report” to receive a structured financial diagnostic covering DTI ratio, risk level, and actionable recommendations.

Technical Architecture:



Description: FinWise AI follows a modular three-tier architecture built for a fully serverless, frontend-only deployment. The presentation tier (HTML, CSS, JavaScript) captures user input across the four tool pages. The business-logic tier — pure vanilla JavaScript running in the browser — applies BFSI business rules, the FICO-style scoring model, and the EMI amortization formula locally, with no dedicated backend server. For AI-driven insights, the logic tier constructs a structured prompt and calls the Claude API (Anthropic) directly from the client; results are stored in localStorage and optionally forwarded to a Google Sheets webhook (Google Apps Script) for persistent, centralized logging.

(Note: FinWise AI does not use a relational database — all financial diagnostic data is stored client-side in browser localStorage, with Google Sheets used only as an optional external log/backup, so no ER diagram is applicable.)

Pre-requisites:

- **HTML5, CSS3 & Vanilla JavaScript (ES6+):** core skills for building the four tool pages and shared UI components.
- **Anthropic Claude API Familiarity:** understanding of the /v1/messages endpoint, request/response structure, and API key management.
- **REST API & JSON Concepts:** for constructing prompts, parsing Claude responses, and handling the Google Sheets webhook payloads.
- **BFSI Domain Concepts:** FICO-style credit scoring (300–900), Debt-to-Income (DTI) ratio, EMI & loan amortization, and credit risk classification.
- **Browser Storage (localStorage):** for persisting the user's diagnostic profile and application settings across pages.
- **Google Apps Script (optional):** for building the serverless webhook that logs data to Google Sheets.
- **Version Control with Git:** for tracking project changes and enabling deployment.
- **Local HTTP Server:** e.g. `python -m http.server 8080` or `npx http-server`, for local testing.

Project Workflow:

Milestone 1: Foundation & Design System

- Activity 1.1: Define the design tokens (colors, typography, spacing) in a shared CSS variables file for the dark glassmorphism theme.
- Activity 1.2: Build the responsive sidebar navigation and shared layout shell used across all five pages.
- Activity 1.3: Set up the project file structure — index.html, eligibility.html, credit.html, emi.html, tips.html, script.js, style.css.
- Activity 1.4: Generate an Anthropic Claude API key and configure secure client-side key management via the in-app Settings modal.

Milestone 2: Loan Eligibility Checker

- Activity 2.1: Design the Eligibility Criteria Form (Applicant Name, Monthly Salary, CIBIL/Credit Score, Existing Monthly EMI, Applicant Age).
- Activity 2.2: Implement the four-rule business logic engine (Salary, Score, EMI, Age thresholds) and the eligible-loan-amount formula.
- Activity 2.3: Build the Diagnostics Ledger panel with a live business-rule checklist and risk-level display.

Milestone 3: Credit Score Analyzer

- Activity 3.1: Design the Credit Risk Calculator form (default history, utilization ratio, credit age, active credit types, hard inquiries).
- Activity 3.2: Implement the FICO-style weighted scoring algorithm across five factors, bounded between 300 and 900.
- Activity 3.3: Build the Credit Diagnostics panel to display the estimated score, risk tier, and contributing factors.

Milestone 4: Quick EMI Calculator

- Activity 4.1: Design the EMI Calculator form (Loan Principal Amount, Annual Interest Rate, Tenure in Months).
- Activity 4.2: Implement the reducing-balance EMI formula and the month-by-month amortization schedule generator.
- Activity 4.3: Build the Payment Schedule panel to render the calculated EMI and ledger table.

Milestone 5: AI Financial Advisor Integration

- Activity 5.1: Integrate the Claude API (claude-3-5-sonnet-20241022) with a custom financial-advisor system prompt.
- Activity 5.2: Build the chat interface, Assessment Context panel, and the one-click “Generate Audit Report” feature.
- Activity 5.3: Implement an offline Demo Mode with simulated AI responses as a fallback when no live API key is configured.

Milestone 6: Data Persistence & Integration

- Activity 6.1: Implement localStorage-based state management for the diagnostic profile and application settings.
- Activity 6.2: Build the optional Google Sheets webhook integration (Google Apps Script) with CSV-download fallback.

Milestone 7: Testing & Local Deployment

- Activity 7.1: Test all four tools end-to-end in both Demo Mode and Live API Mode.
- Activity 7.2: Serve the application locally via a lightweight HTTP server and verify responsiveness across breakpoints.

Milestone 8: Conclusion

Milestone 1: Foundation & Design System

This milestone establishes the visual identity and shared infrastructure of FinWise AI — a dark, glassmorphism-styled BFSI interface built with the Outfit and Inter typefaces, an electric-blue accent color, and a consistent card-based layout reused across all five pages.

Activity 1.1: Define the Design Token System

A central CSS variables file establishes the visual language of the platform, including the dark background palette, glass-card blur effects, and accent colors used throughout the UI.

```
:root {
  --font-display: 'Outfit', -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto,
  sans-serif;
  --font-body: 'Inter', -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-
  serif;

  /* Colors */
  --bg-primary: #030712;
  --bg-secondary: #0a0d16;
  --accent-blue: #3b82f6;
}
```

Activity 1.2: Generate the Claude API Key

- Step 1 — Create an Anthropic Account: visit <https://console.anthropic.com> and sign up.
- Step 2 — Navigate to the API Keys section of the console and generate a new key.
- Step 3 — Store the key securely via the in-app Settings modal, which persists it to localStorage under appSettings.apiKey — never hard-code the key directly in script.js.
- Step 4 — Toggle between Demo Mode (offline, no key required) and Live Mode (calls the real Claude API) from the same Settings modal.

Activity 1.3: Initialize Application State

```
// Application Configuration State
let appSettings = {
  mode: 'demo', // 'demo' or 'live'
  apiKey: '',
  sheetUrl: ''
};
let currentAnalysis = null;

// Initialize on DOM load
window.addEventListener('DOMContentLoaded', () => {
  // 1. Load configuration settings
  const savedSettings = localStorage.getItem('finewise_settings');
```

```
    if (savedSettings) appSettings = JSON.parse(savedSettings);  
  });
```

Milestone 2: Loan Eligibility Checker

This milestone implements the core rule-based loan eligibility engine — the first of FinWise AI's four financial tools.

Activity 2.1: Capture Applicant Details

- Applicant Name, Monthly Salary (₹), CIBIL / Credit Score (300–900), Existing Monthly EMI (₹), and Applicant Age are collected via the Eligibility Criteria Form.

Activity 2.2: Apply BFSI Business Rules

Eligibility is determined using four independent business rules, all of which must pass (AND logic):

```
function checkEligibility(event) {
  if (event && event.preventDefault) event.preventDefault();

  const salary = parseFloat(document.getElementById('salary').value);
  const score = parseInt(document.getElementById('score').value);
  const emi = parseFloat(document.getElementById('emiInput').value);
  const age = parseInt(document.getElementById('age').value);

  // Business rules
  const rSalaryPass = salary >= 20000;
  const rScorePass = score >= 600;
  const rEmiPass = emi < 20000;
  const rAgePass = age >= 21;

  const allRulesPass = rSalaryPass && rScorePass && rEmiPass && rAgePass;
  const eligibleAmount = allRulesPass ? (salary * 20) : 0;

  currentAnalysis = {
    name, salary, creditScore: score, emiInput: emi, age,
    amount: eligibleAmount, status: allRulesPass ? 'Approved' : 'Rejected'
  };
  localStorage.setItem('finewise_analysis', JSON.stringify(currentAnalysis));
}
```

Activity 2.3: Classify Risk & Render the Diagnostics Ledger

```
if (score >= 800) riskCategory = 'Very Low';
else if (score >= 720) riskCategory = 'Low';
else if (score >= 650) riskCategory = 'Moderate';
else if (score >= 580) riskCategory = 'High';
else riskCategory = 'Very High';
```


The Diagnostics Ledger panel displays the final Approved/Rejected status, the calculated risk level, the eligible loan amount, and a pass/fail checklist for all four business rules, giving the applicant full transparency into the decision.

Milestone 3: Credit Score Analyzer

This milestone implements a FICO-style weighted credit scoring model that estimates an applicant's creditworthiness on a 300–900 scale using five weighted factors.

Activity 3.1: Weighted Scoring Factors

Factor	Weight	Max Points	Rule
Payment History	35%	210 pts	No defaults → +210, else +50
Credit Utilization	30%	180 pts	≤15% → +180 ... else +15 (tiered)
Credit Age	15%	90 pts	≥10 yrs → +90 ... else +5 (tiered)
Hard Inquiries	10%	60 pts	0 → +60, ≤2 → +40, ≤4 → +15, else +0
Credit Diversity	10%	60 pts	≥3 types → +60, ≥1 → +40, else +10

Activity 3.2: Implement the Scoring Engine

```
function analyzeCredit(event) {
  event.preventDefault();

  let score = 300; // Base score

  // Payment History (35%)
  score += (defaultFlag === 'no') ? 210 : 50;

  // Credit Utilization (30%)
  if (utilization <= 15) score += 180;
  else if (utilization <= 30) score += 145;
  else if (utilization <= 50) score += 100;
  else if (utilization <= 75) score += 50;
  else score += 15;

  // Credit Age (15%)
  if (age >= 10) score += 90;
  else if (age >= 5) score += 70;
  else if (age >= 3) score += 50;
  else if (age >= 1) score += 20;
  else score += 5;

  // Hard Inquiries (10%) & Credit Diversity (10%) applied similarly...
```

```
score = Math.min(900, Math.max(300, score));  
localStorage.setItem('elected_credit_score', score);  
}
```

Activity 3.3: Risk Tier Classification

- Excellent: 750–900 — Very Low Risk
- Good: 650–749 — Moderate Risk
- Poor: 300–649 — High Risk

Milestone 4: Quick EMI Calculator

This milestone implements the reducing-balance EMI formula used across the BFSI industry to compute fixed monthly loan installments and a full amortization schedule.

Activity 4.1: EMI Formula

$$EMI = [P \times R \times (1 + R)^N] \div [(1 + R)^N - 1]$$

- P = Principal loan amount
- R = Monthly interest rate (Annual Rate ÷ 12 ÷ 100)
- N = Loan tenure in months

Activity 4.2: Implement the Calculation Engine

```
const defCalculateEmi = (principal, annualRate, tenureMonths) => {  
  const r = (annualRate / 12) / 100; // Monthly interest rate  
  const n = tenureMonths;  
  
  if (r === 0) return principal / n;  
  
  const emi = (principal * r * Math.pow(1 + r, n)) / (Math.pow(1 + r, n) - 1);  
  return emi;  
};
```

Activity 4.3: Generate the Amortization Table

```
let balance = principal;  
const mr = (rate / 12) / 100;  
  
for (let i = 1; i <= tenure; i++) {  
  const interest = balance * mr;  
  const principal_paid = emi - interest;  
  balance -= principal_paid;  
  // Append row: Month | EMI | Interest | Principal | Balance  
}
```

Example: a ₹5,00,000 loan at 10.5% annual interest over 60 months computes to an EMI of approximately ₹10,746.95 per month, with the full month-wise breakdown rendered in the Payment Schedule panel.

Milestone 5: AI Financial Advisor Integration

This milestone integrates Anthropic's Claude API to deliver conversational, context-aware financial coaching based on the user's live diagnostic profile.

Activity 5.1: Claude API Configuration

- **Model:** claude-3-5-sonnet-20241022
- **max_tokens:** 1024
- **API Version Header:** anthropic-version: 2023-06-01
- **System Prompt:** a custom financial-advisor persona instructing Claude to act as a professional credit risk underwriter and financial planner.

```
const response = await fetch('https://api.anthropic.com/v1/messages', {
  method: 'POST',
  headers: {
    'x-api-key': apiKey,
    'anthropic-version': '2023-06-01',
    'content-type': 'application/json'
  },
  body: JSON.stringify({
    model: 'claude-3-5-sonnet-20241022',
    max_tokens: 1024,
    messages: [{ role: 'user', content: prompt }],
    system: systemPrompt
  })
});
```

Activity 5.2: Build the Chat & Audit Report Interface

- A real-time chat panel lets the user ask free-form financial questions.
- A one-click “Generate Audit Report” button sends the full diagnostic context (income, DTI ratio, credit score, requested loan amount) to Claude for a structured financial audit.
- The Assessment Context sidebar displays the Calculated Risk Level, Eligibility Status, Monthly Income, DTI Ratio, Credit Score, and Requested Loan Amount in real time.

Activity 5.3: Offline Demo Mode Fallback

When no live API key is configured, FinWise AI automatically falls back to an offline Demo Mode that simulates AI responses locally, ensuring the platform remains fully functional for demonstration and testing purposes.

Milestone 6: Data Persistence & Google Sheets Integration

Activity 6.1: LocalStorage State Management

```
// Main analysis profile
localStorage.setItem('finewise_analysis', JSON.stringify({
  name, salary, creditScore, emiInput, age,
  amount, riskCategory, status, reason, dti
}));

// App settings
localStorage.setItem('finewise_settings', JSON.stringify({
  mode: 'demo' | 'live',
  apiKey: '...',
  sheetUrl: '...'
}));
```

Activity 6.2: Google Sheets Webhook (Google Apps Script)

Users may optionally configure a Google Apps Script webhook URL via the Settings modal. On each eligibility check, the application POSTs the applicant's key details to the webhook, which appends a new row to a connected Google Sheet — giving teams a lightweight, serverless data log without needing a traditional database.

```
function doPost(e) {
  const sheet = SpreadsheetApp.getActiveSheet();
  const data = JSON.parse(e.postData.contents);

  sheet.appendRow([data.name, data.salary, data.score, data.emi, data.result]);

  return ContentService.createTextOutput(JSON.stringify({status: 'success'}));
}
```

If the webhook call fails or no sheet URL is configured, the application falls back to a local CSV download so no diagnostic data is lost.

Milestone 7: Testing & Local Deployment

Activity 7.1: Local Development Server

- Python 3: `python -m http.server 8080`
- Node.js: `npx http-server -p 8080`

Once running, the application is accessed locally at `http://localhost:8080`.

Activity 7.2: Functional Testing Checklist

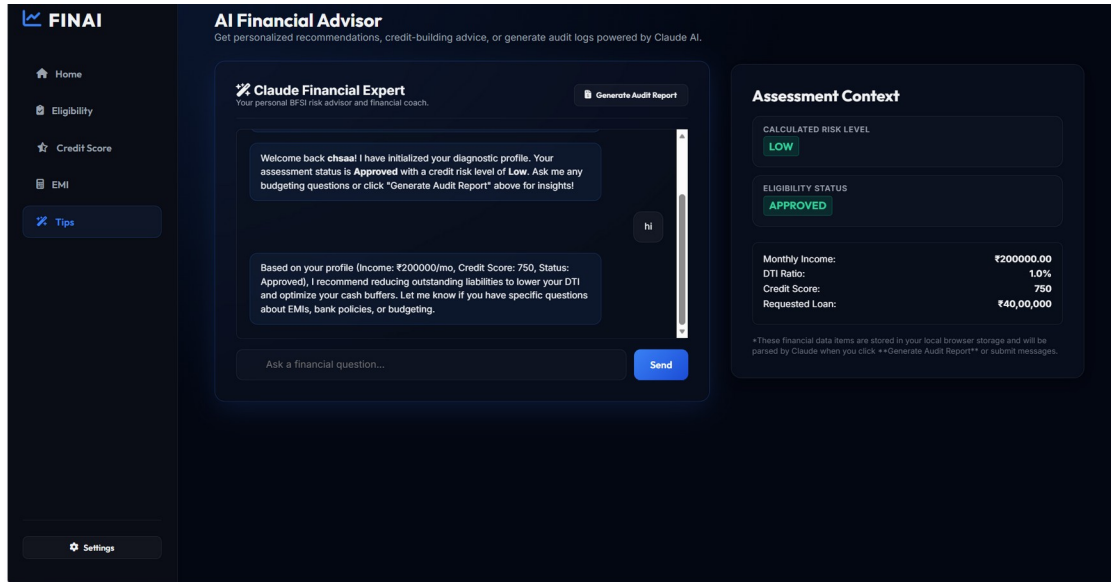
- Verify all four business rules on the Eligibility Checker across passing and failing input combinations.
- Validate the Credit Score Analyzer against known score ranges (Excellent / Good / Poor).
- Cross-check EMI Calculator output and amortization table against manual reducing-balance calculations.
- Test the AI Financial Advisor in both Demo Mode and Live API Mode, confirming the Assessment Context updates correctly.
- Confirm responsive behavior across desktop, tablet, and mobile breakpoints.

Important Notes

- The Claude API key is stored only in the browser's localStorage and is never committed to source control.
- Live Mode requires a valid Anthropic API key configured via the Settings modal; without one, the app automatically operates in Demo Mode.
- The Google Sheets webhook URL is optional — the platform is fully functional without it, using localStorage and CSV export as defaults.

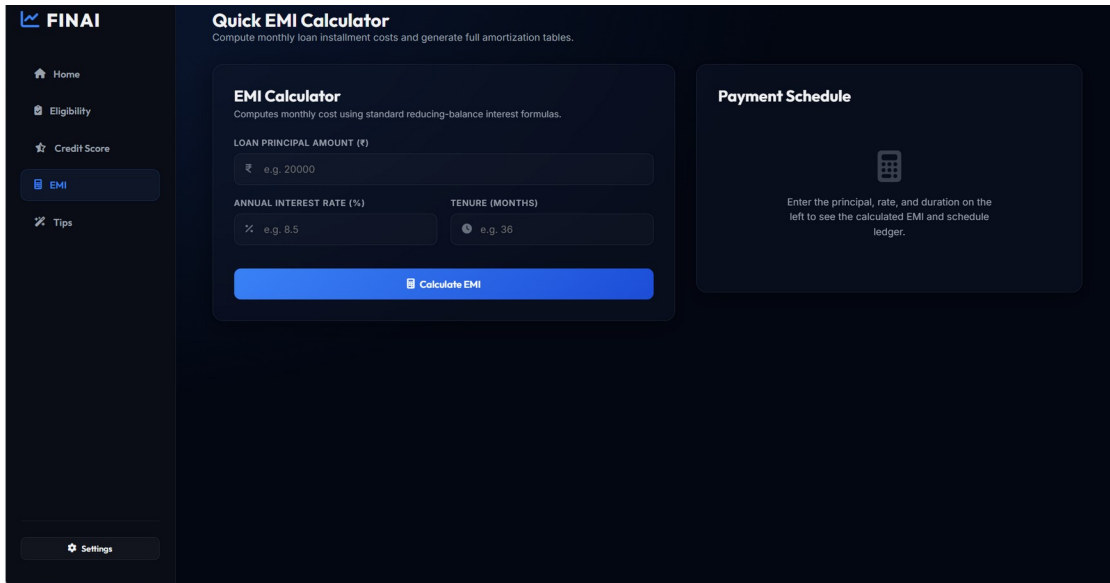
Exploring the Website's Web Pages:

Home Page:



Description: The landing page introduces the AI Loan Eligibility Checker platform with a clear headline, a “Get Started” call-to-action, and three highlighted capability cards — Multi-Factor Diagnostics, EMI Amortization, and Claude AI Financial Coaching — followed by an AI Financial Tips section covering Smart Savings, Improve Credit Score, and Budget Management.

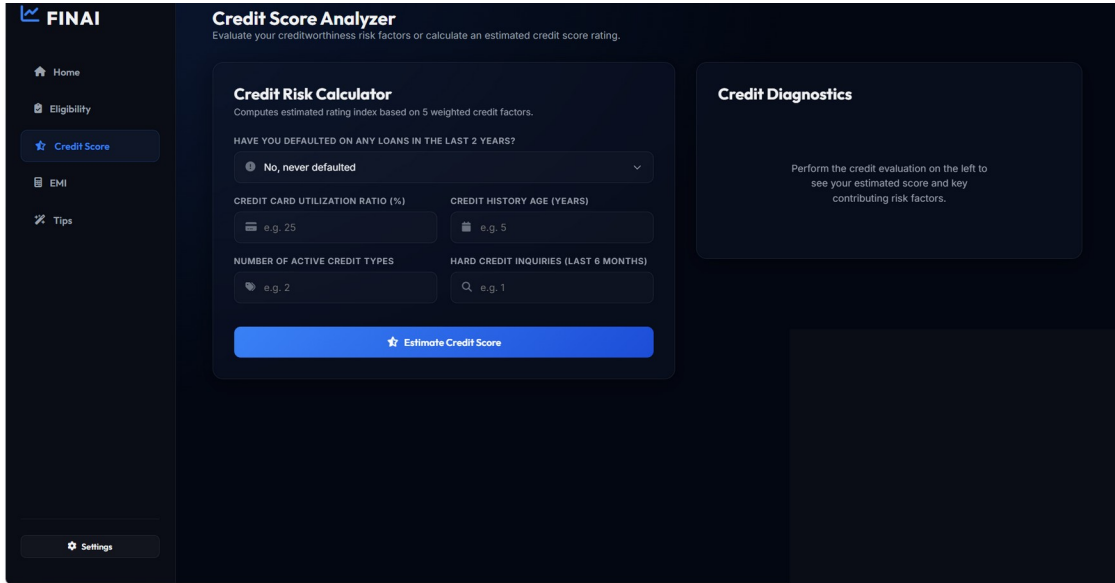
Loan Eligibility Checker Page:



The screenshot shows a web application interface for a loan eligibility checker. On the left is a dark sidebar with the 'FINAI' logo and navigation links: Home, Eligibility, Credit Score, EMI (highlighted), Tips, and Settings. The main content area has a dark background. At the top, it says 'Quick EMI Calculator' with a subtext 'Compute monthly loan installment costs and generate full amortization tables.' Below this are two panels. The 'EMI Calculator' panel on the left contains input fields for 'LOAN PRINCIPAL AMOUNT (₹)' (with a hint 'e.g. 20000'), 'ANNUAL INTEREST RATE (%)' (with a hint 'e.g. 8.5'), and 'TENURE (MONTHS)' (with a hint 'e.g. 36'). A blue 'Calculate EMI' button is at the bottom. The 'Payment Schedule' panel on the right has a calculator icon and text: 'Enter the principal, rate, and duration on the left to see the calculated EMI and schedule ledger.'

Description: A glass-styled two-column layout. The left panel contains the Eligibility Criteria Form (Applicant Name, Monthly Salary, CIBIL/Credit Score, Existing Monthly EMI, Applicant Age). The right panel is the Diagnostics Ledger, which displays the Approved/Rejected verdict, risk level, eligible loan amount, a pass/fail checklist for all four business rules, and a “Save to Sheets” action.

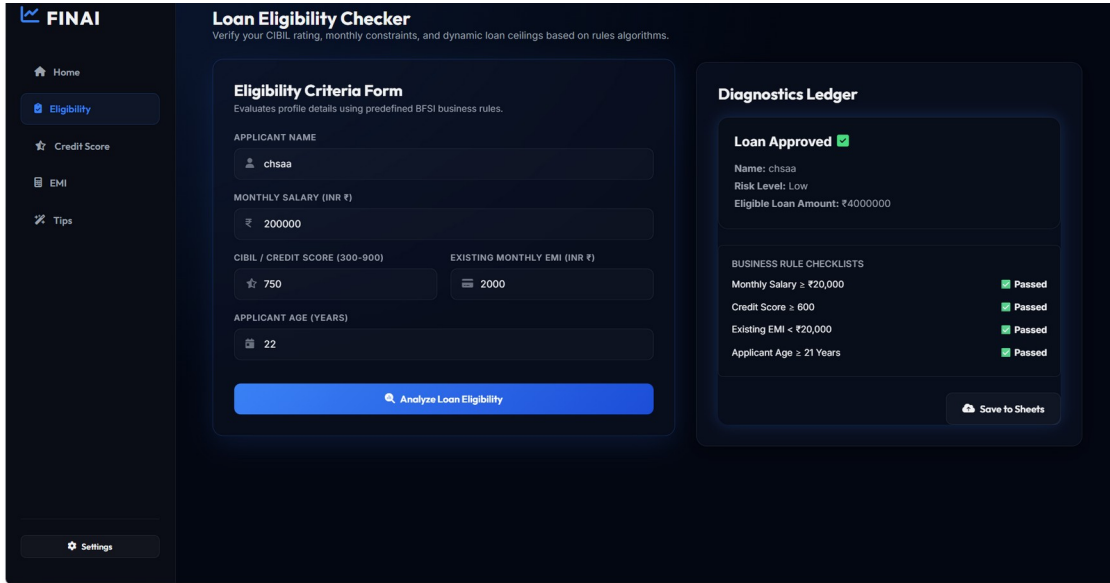
Credit Score Analyzer Page:



The screenshot shows the 'Credit Score Analyzer' page. On the left is a dark sidebar with the 'FINAL' logo at the top and navigation links: Home, Eligibility, Credit Score (highlighted with a star), EMI, Tips, and Settings. The main content area has a title 'Credit Score Analyzer' and a subtitle 'Evaluate your creditworthiness risk factors or calculate an estimated credit score rating.' Below this is the 'Credit Risk Calculator' section, which states it 'Computes estimated rating index based on 5 weighted credit factors.' It contains five input fields: 'HAVE YOU DEFAULTED ON ANY LOANS IN THE LAST 2 YEARS?' (a dropdown menu showing 'No, never defaulted'), 'CREDIT CARD UTILIZATION RATIO (%)' (input field with 'e.g. 25'), 'CREDIT HISTORY AGE (YEARS)' (input field with 'e.g. 5'), 'NUMBER OF ACTIVE CREDIT TYPES' (input field with 'e.g. 2'), and 'HARD CREDIT INQUIRIES (LAST 6 MONTHS)' (input field with 'e.g. 1'). A large blue button labeled 'Estimate Credit Score' is at the bottom of this section. To the right is a 'Credit Diagnostics' panel with the instruction 'Perform the credit evaluation on the left to see your estimated score and key contributing risk factors.' Below this panel is a large, empty rectangular box.

Description: The Credit Risk Calculator form collects default history, credit utilization ratio, credit history age, number of active credit types, and hard inquiries in the last six months. The Credit Diagnostics panel on the right renders the estimated score and key contributing risk factors once the evaluation is run.

Quick EMI Calculator Page:

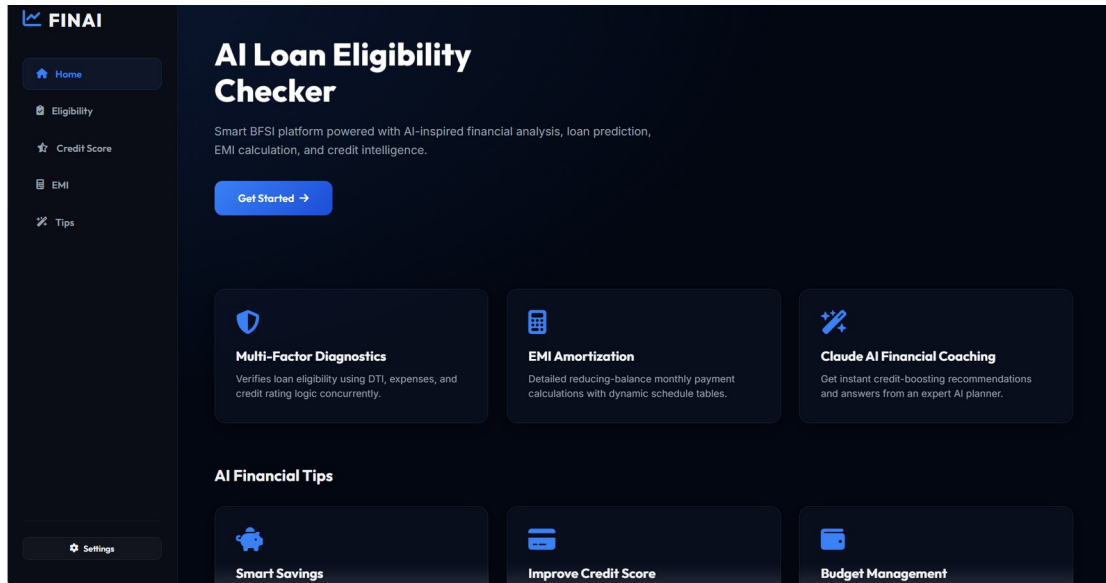


The screenshot shows a web application interface for a 'Loan Eligibility Checker'. On the left is a dark sidebar with a 'FINAL' logo and navigation links: Home, Eligibility (active), Credit Score, EMI, Tips, and Settings. The main content area is titled 'Loan Eligibility Checker' with a subtitle 'Verify your CIBIL rating, monthly constraints, and dynamic loan ceilings based on rules algorithms.' It is divided into two panels. The left panel, 'Eligibility Criteria Form', contains input fields for 'APPLICANT NAME' (chsa), 'MONTHLY SALARY (INR ₹)' (₹ 200000), 'CIBIL / CREDIT SCORE (300-900)' (750), 'EXISTING MONTHLY EMI (INR ₹)' (2000), and 'APPLICANT AGE (YEARS)' (22). A blue button at the bottom says 'Analyze Loan Eligibility'. The right panel, 'Diagnostics Ledger', shows a 'Loan Approved' status with a green checkmark. It lists details: Name: chsa, Risk Level: Low, and Eligible Loan Amount: ₹4000000. Below this is a 'BUSINESS RULE CHECKLISTS' table with four rows, each showing a condition and a 'Passed' status with a green checkmark. A 'Save to Sheets' button is at the bottom right of this panel.

BUSINESS RULE CHECKLISTS	
Monthly Salary > ₹20,000	Passed
Credit Score > 600	Passed
Existing EMI < ₹20,000	Passed
Applicant Age > 21 Years	Passed

Description: Users enter the Loan Principal Amount, Annual Interest Rate, and Tenure (Months) into the EMI Calculator card. The Payment Schedule panel on the right renders the calculated monthly EMI and the full amortization ledger once the calculation is run.

AI Financial Advisor Page:



Description: The Claude Financial Expert chat panel greets the user with a summary of their current diagnostic status and responds to free-form financial questions. The Assessment Context sidebar displays the Calculated Risk Level, Eligibility Status, Monthly Income, DTI Ratio, Credit Score, and Requested Loan Amount, all sourced from the user's local diagnostic profile.

Skills Required

- HTML5
- CSS Animations
- JavaScript (Programming Language)
- Generative Artificial Intelligence
- REST APIs
- JSON
- Git (Version Control System)

Team Members

- Karthikeya Girindra Animella — Team Lead
- Meduri Prem Sai — Member

Technology Stack Summary

Aspect	Details
AI Model	Claude Sonnet 3.5 (claude-3-5-sonnet-20241022) via Anthropic API
Architecture	Frontend-only (vanilla JS) with optional serverless Google Apps Script webhook
Backend	None — fully client-side
Data Storage	Browser localStorage, with optional Google Sheets webhook and CSV export fallback
Core Tools	Loan Eligibility Checker, Credit Score Analyzer, EMI Calculator, AI Financial Advisor
Tech Stack	HTML5, CSS3, Vanilla JavaScript, Anthropic Claude API
UI Style	Dark glassmorphism theme, fully responsive

Conclusion:

FinWise AI is a serverless, browser-native BFSI platform that brings together rule-based loan eligibility assessment, FICO-style credit scoring, reducing-balance EMI calculation, and Claude-powered financial coaching in a single, cohesive glassmorphism interface. By running all core business logic in vanilla JavaScript and calling the Claude API directly from the client — with localStorage and an optional Google Sheets webhook for persistence — the project demonstrates how targeted generative AI and a well-structured rules engine can deliver a fast, dependency-light financial decision-support tool without a traditional backend.

Looking ahead, FinWise AI is well positioned for further growth. Planned enhancements include real-time banking API integration for live salary and account verification, user authentication for multi-user profiles, advanced risk dashboards with historical trend visualizations, predictive credit-risk modeling using machine learning, and native mobile application support. Continued refinement of the underlying Claude prompts and expansion of the API integration layer will help the platform evolve from a focused diagnostic tool into a comprehensive personal financial management assistant.